Price Signal

Technical report - ModelX 2.0, Phase 2: Implementation - Domain 01: Cost of living and everyday economics

Author: Nivetha Ramanan (SLIIT) - Repository: github.com/IT25101414/price-signal - Data: Central Bank of Sri Lanka Daily Price Reports, 28 July 2020 to 15 October 2026 - Revised 18 October 2026

1. Summary

The problem. Small food buyers in Colombo cannot tell whether an unusual wholesale move at Dambulla is likely to reach Pettah or Narahenpita. The Central Bank publishes prices each working day, but its PDF is a price table rather than a forward-looking buying aid.

What we built. A respectful PDF collector, position-based table parser, time-ordered ridge model, split-conformal 80% uncertainty range, explicit safety guard, six-page Flask/SQLite website and a secondary Streamlit dashboard. Welcome introduces the independent project; Today summarizes source health and guarded estimates; Explore compares paired history; Weekly report compares complete five-report-day windows; How it works presents evidence and limits; Help explains the public-facing labels.

What we found. On held-out 2024-2026 data, adding matching Dambulla information improved three-day forecasts beyond an own-history model by 9.7 points for wholesale Pettah, 8.3 for retail Pettah and 12.1 for retail Narahenpita. Three-day 80% ranges contained 81.0%, 81.5% and 77.4% of outcomes. A decision simulation was 1.22% to 1.57% cheaper than always buying today; these are modelling results, not observed buyer savings.

Independent checks. Three non-overlapping 2026 rolling windows produced positive Dambulla gains in all 9 market-window combinations (+5.5 to +13.7 points), while all 9 deliberately wrong-item placebos were zero or negative. The gradient-boosting challenger did not satisfy the pre-declared adoption rule, so the simpler ridge model remains.

People and safety. Three consented small-food buyers completed formative sessions on 16-17 October. Median understanding, trust as one input and usefulness were each 4/5. Their confusion about the possible range, shared report days and blank weekly values directly changed the interface wording. All 86 repository-owned tests pass with zero environment-dependent skips, and all 12 deliberate refusal cases behave as specified.

Current status. The implementation, report, proposal, two-page summary, MIT licence, AI disclosure and redacted CBSL reply are present. Public deployment verification, the three-minute video and a final signed-out browser rehearsal remain pending.

2. The proposal against what exists

What the proposal promised	Status	Evidence or reason
Respect robots.txt and 10-second delay; keep a manifest	Complete	download_cbsl_reports.py; 2,274 records discovered, 2,273 downloaded
Position-based parser; reject unknown layouts	Complete	extract_prices.py; 1,487 supported report days; failed_files.txt
Lead-lag evidence, intervals and placebo	Complete	leadlag_check.py, robust_check.py and saved results
Decision-value backtest	Complete as simulation	decision_backtest.py; assumptions stated
Calibrated 80% uncertainty range	Complete	77.4%-81.5% three-day coverage; 75%-85% target met
Safe refusal for stale, incomplete or weak evidence	Complete	guard.py; 12/12 constructed refusal cases pass
Daily refresh	Complete locally	refresh.py and run_daily.sh; cloud schedule deliberately deferred
Public-facing interface	Complete locally	Welcome, Today, Explore, Weekly report, How it works and Help
Past-price charts and weekly comparison	Complete	Paired charts, guarded CSV export and print-to-PDF
Evidence page tied to saved results	Complete	web/evidence.py reads repository result files
Automated tests on every push	Complete	GitHub Actions; 86 passed, zero skipped locally
Two or three buyer sessions	Complete	Three consented sessions; individual ratings and difficulties reported
Gradient-boosting challenger	Complete, not adopted	Did not pass the pre-declared rule
Written data-use clarification	Received	CBSL reply dated 12 October 2026; redacted copy committed
Sinhala and Tamil interface	Not implemented	Lower priority; must not delay reliable final submission
Public deployment and video	Pending	Owner-authorized deployment, URL verification and recording required

3. Architecture

Data moves in one direction and no estimate reaches a user without passing the safety guard. The same guard runs when cards are created and again on every website request, because a previously safe card can later become stale.

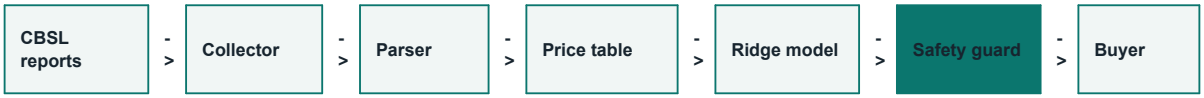


Figure 1. The end-to-end pipeline. The safety guard is the only route from the model to the public interface.

3.1 Operating sequence

- The collector reads the official listing, obeys the crawl delay, downloads each report once and records URL, report date, hash, status and time.

- The parser reads words by horizontal position, preserves missing and closed-market states, rejects unknown layouts and never guesses a shifted value.

- The model compares a Colombo own-history baseline with the same model plus matching Dambulla information, then produces a three-report-day estimate and an 80% possible range.

- The guard removes output when source health, freshness, current prices, paired history or signal strength are inadequate.

- The Flask site rebuilds its SQLite view from the committed cards and one-year public history slice; the complete generated table and source PDFs stay local.

3.2 Components

Part	What it does
download_cbsl_reports.py	Reads the listing, respects robots.txt and the ten-second delay, downloads each PDF once and writes the manifest.
extract_prices.py	Reads tables by word position. Rejects unknown layouts and records failures instead of shifting values between markets.
refresh.py	Checks the newest listing, appends accepted reports once and logs ok, no_new_report, behind, partial, failed or source_unavailable.
guard.py	Refuses stale data, an unread newer report, closures, missing prices, insufficient paired history and weak signals.
model.py	Fits pooled ridge models and split-conformal 80% ranges; every public card passes through the guard.
web/	Flask/SQLite server plus six responsive pages. Guarded APIs remove unsafe values server-side, not just visually.
export_recent.py	Writes the one-year history slice needed for public charts without redistributing the full generated table.
rolling_stress_check.py	Refits the model before three later 2026 windows and compares the real Dambulla signal with a wrong-item placebo.
accessibility_check.py	Checks headings, accessible names, table alternatives and light/dark contrast without external services.
smoke_test.py	Checks /healthz, /api/status and every public page against a local or deployed URL.
challenger_check.py	Tests a gradient-boosting alternative under a decision rule fixed before evaluation.
app.py	Retains the guarded Streamlit dashboard as a secondary interface.

4. Technology and why

Part	Choice	Why
Language	Python 3.9+	One language for collection, parsing, modelling, tests and interfaces.
Download	requests, urllib.robotparser	Explicit pacing, identifiable User-Agent and fail-closed robots handling.
PDF parsing	pdfplumber	Exposes word positions needed when empty PDF cells disappear from extracted text.
Modelling	NumPy ridge regression	Few parameters, stable under collinearity and explainable coefficients.
Data	pandas, CSV, SQLite	CSV remains reproducible source; SQLite serves item-level queries efficiently.
Interface	Flask; plain HTML/CSS/JS; Streamlit	No CDN or build step; responsive charts, text alternatives and light/dark themes.
Testing	pytest; GitHub Actions	86 reproducible tests plus saved refusal, rolling and accessibility evidence.
Licence	MIT for original software	Keeps code open while third-party data, reports and emblem retain their own terms.

5. The data finally used

5.1 Source and access

The source is the Statistics Department, Central Bank of Sri Lanka, Daily Price Report. Each current PDF lists wholesale and retail prices for food items and markets including Pettah, Dambulla and Narahenpita. The source contains prices by item, market and report date, not personal data.

The collector uses only the official listing and report files. It respects the published crawl delay, keeps raw files unchanged locally and records hashes. The repository does not redistribute the raw PDF archive or the complete generated price table.

5.2 What we obtained

Measure	Verified position at 18 October 2026
Manifest records	2,274 discovered through the 15 October report
Downloads	2,273 successful; one recorded failure
Supported analysis period	1,487 report days from 28 July 2020
Public build	Cards and a one-year history slice through 15 October 2026
Extractor verification	Repository tests plus a separate 125-value check against local source PDFs

5.3 Cleaning and source irregularities

Layouts change, scans may have no usable text layer, report headers can be undated, comparison dates vary and item names drift. Blank, 'n.a.' and market-closed cells remain distinct. Exact duplicate files are rejected. A disagreement between a report summary and its own table is retained as a documented source issue; the table is the analysis input.

5.4 Right to use, attribution and ethics

CBSL's Statistics Department replied in writing on 12 October 2026 that the published data are publicly available and do not require specific approval for use, requested attribution and stated that CBSL does not endorse or accept responsibility for derived findings. A one-page redacted copy is committed as docs/cbsl_reply_2026-10-12.pdf. Every public page carries attribution and non-endorsement wording.

6. Temporality

Historical and current data serve different purposes. Years of ordered history train and test the relationship; the newest accepted report drives today's public view. refresh.py checks for newer reports and logs every run. If the official listing shows an unread report, or if the last good report is more than three working days old, the interface removes estimates and explains why.

Working-day freshness excludes weekends and dates listed in data/holidays.csv. The file currently covers 2026 only, so it must be extended before later operational use. Missing a holiday makes the guard refuse too early, which is the safer failure direction.

7. Method

Target. The log change of a Colombo series over the next one, three or five report days. The public product uses three report days.

Models. Model C uses the Colombo series' recent one-day and five-day changes and its distance from its own 20-day mean. Model D adds matching Dambulla one-day and five-day changes and the Dambulla-minus-Colombo gap relative to its trailing 60-day average. The reported gain is D beyond C, not performance against an intentionally weak baseline.

Time ordering. Evaluation models fit data through 2023 and test from 2024 onward. Labels that cross the split are removed. Robustness intervals resample complete months, retaining local dependence.

Uncertainty. The 80% range is split-conformal: held-out residuals are scaled by recent item volatility before the quantile is selected. This gives an empirically checked possible range, not a guarantee or a best/worst-case bound.

Signal labels. Thresholds are fixed from training data. The website says 'no clear signal' when evidence is too weak and never replaces a refusal with a cosmetic warning.

Safety guard. A card is removed for a stale report, unread newer report, relevant market closure, missing Dambulla or Colombo price, fewer than 40 paired observations among the last 60 report days, or a gap too small to support the model's historical skill.

8. Evaluation against the proposal

The main evaluation uses held-out 2024-2026 data. Later checks use the committed one-year history slice and three chronological, non-overlapping 2026 windows. All results are saved as repository text files or reproducible scripts.

The following questions are evaluated separately: Does Dambulla add forecast value beyond own history? Is the stated uncertainty calibrated? Does a simulated timing policy improve on always buying today? Does the system refuse in every declared unsafe state? Are the findings stable across later windows and absent under a wrong-item placebo?

8.1 Forecast value: does Dambulla add anything?

Colombo series	3-day gain	Held-out interval	Wrong-item placebo	Items gaining
Wholesale Pettah	+9.7 pts	above zero	approximately zero	majority
Retail Pettah	+8.3 pts	above zero	approximately zero	majority
Retail Narahenpita	+12.1 pts	above zero	approximately zero	majority

Gain is the share of squared forecast error removed beyond the own-history model. The placebo substitutes another item's Dambulla series and should show no skill.

8.2 Calibration: is the range honest?

Colombo series	Test cases	Inside stated 80% range	Average half-width
Wholesale Pettah	10,407	81.0%	+/- 28.1%
Retail Pettah	10,739	81.5%	+/- 23.3%
Retail Narahenpita	10,129	77.4%	+/- 18.1%

All three overall values fall inside the proposal's 75%-85% acceptance band. The ranges are intentionally wide because short-horizon food prices remain noisy.

8.3 Decision value: is it useful in a timing simulation?

Colombo series	Dambulla policy vs buy today	Own-history policy	Hindsight ceiling
Wholesale Pettah	-1.42%	+0.15%	-5.34%
Retail Pettah	-1.22%	-0.06%	-4.40%
Retail Narahenpita	-1.57%	-0.52%	-3.56%

Negative means cheaper. The simulated buyer chooses today or three report days later using thresholds fixed before the test. This assumes that deferral has no shortage, spoilage, transport or operating cost; it is not observed buyer savings.

8.4 Safe failure

refusal_report.py creates synthetic tables with exactly one fault each. These test logic, not real-market prevalence.

Declared failure	Result	Public behavior
Report more than 3 working days old	Pass	All estimates removed; stale reason shown
Newer official report listed but unread	Pass	Previous cards refused
Dambulla or selected Colombo market closed	Pass	Affected output removed
Missing or n.a. source price	Pass	No estimated value shown
Too little recent paired history	Pass	Insufficient-data reason shown
Gap too small to support evidence	Pass	No-clear-signal state shown
Source listing unavailable	Pass	Refresh fails closed and records status
Withheld API card leaks a number	Pass	Server removes protected fields
Interface says buy or sell	Pass	Tests scan rendered output

8.5 Extraction accuracy

Repository-owned unit tests cover cell parsing, row grouping, name normalization and cross-report consistency without external fixtures. A separate 125-value validation compares extracted values with original CBSL PDFs locally because those source reports are not redistributed.

8.6 The proposal's own failure tests

Proposal failure condition	Triggered?	Evidence
Most gain intervals include zero	No	Main held-out gains are above zero
Placebo reaches half the real gain	No	Wrong-item placebos are approximately zero
80% range covers below 70% or above 90%	No	77.4%-81.5% at three days
Buyer cannot explain the main card	Partly	One of three needed help; wording revised
Any refusal case fails	No	12/12 behave as declared

8.7 What did not work

Jump persistence. A first attempt predicted whether a one-day jump would persist. Accuracy and AUC did not beat the majority baseline, so it was rejected and retained only as a documented negative experiment.

Gradient boosting. A histogram gradient-boosting challenger used the same features, split and scoring under a pre-declared adoption rule. It was worse than ridge in seven of nine horizon-series comparisons and clearly worse at three days for retail Pettah and retail Narahenpita. Ridge remains because the challenger did not earn the added complexity.

8.8 Rolling-origin stress check

Each model is refitted using only earlier rows before three non-overlapping 2026 windows. Entries are Dambulla-enhanced squared-error gains beyond the own-history model, in percentage points.

Window	Test period	Wholesale	Retail Pettah	Retail Narahenpita	Placebo
W1	5 May-26 Jun	+5.5	+6.0	+12.1	all <= 0
W2	30 Jun-21 Aug	+10.0	+8.5	+10.8	all <= 0
W3	24 Aug-15 Oct	+13.7	+12.9	+12.7	all <= 0

All 9 real gains are positive (+5.5 to +13.7 points), all 9 wrong-item placebos are zero or negative, and a majority of items benefit in every market-window combination. This strengthens recent temporal stability, but the available one-year slice starts in October 2025 and therefore does not test the 2022 crisis out of sample.

8.9 Limits of the evidence

The effect is modest, prices are rounded, report-day horizons cross weekends and holidays, only 18-19 foods have a Dambulla wholesale series, and buyer sessions measure short-task comprehension and intent rather than long-term adoption, changed purchasing behavior or realized savings.

9. Gaps and limitations

Gap	Effect and current response
Public deployment not yet verified	Local build and deployment files are ready; exact public URL must be verified before submission.
Revised wording not retested	Buyer feedback caused changes, but a short follow-up with one or two buyers would confirm comprehension.
Full table and raw PDFs not in repository	Reviewers can rebuild them; committed cards and one-year slice keep the public demo reproducible.
Older layouts excluded	Analysis begins 28 July 2020 instead of guessing incompatible tables.
Supported-layout exceptions remain	Each is listed with its reason; no value is silently repaired.
Many cards intentionally withheld	Coverage is lower, but weak or incomplete evidence cannot appear as a polished number.
Holiday file covers 2026	Extend before later operational use; missing dates cause conservative refusal.
Daily publication is manual	Avoids silently republishing the complete derived table without an explicit decision.
Simulation assumptions	No observed savings claim; supplier confirmation remains necessary.

10. What the user sees

Welcome states who the tool is for and that it is an independent student project. Today shows report date and source health, availability counts, expected direction, possible-range width, a comparison chart, strongest-signal list, filters, guarded export and item details. Explore presents paired price history and a professional Colombo-Dambulla difference summary. Weekly report compares two complete five-report-day windows and explains blank values. How it works presents evaluation evidence, important limits and data provenance. Help explains every label and refusal in plain language.

All pages attribute the Statistics Department, state that CBSL does not endorse the project and describe estimates as uncertain information rather than purchasing advice.

10.1 Today

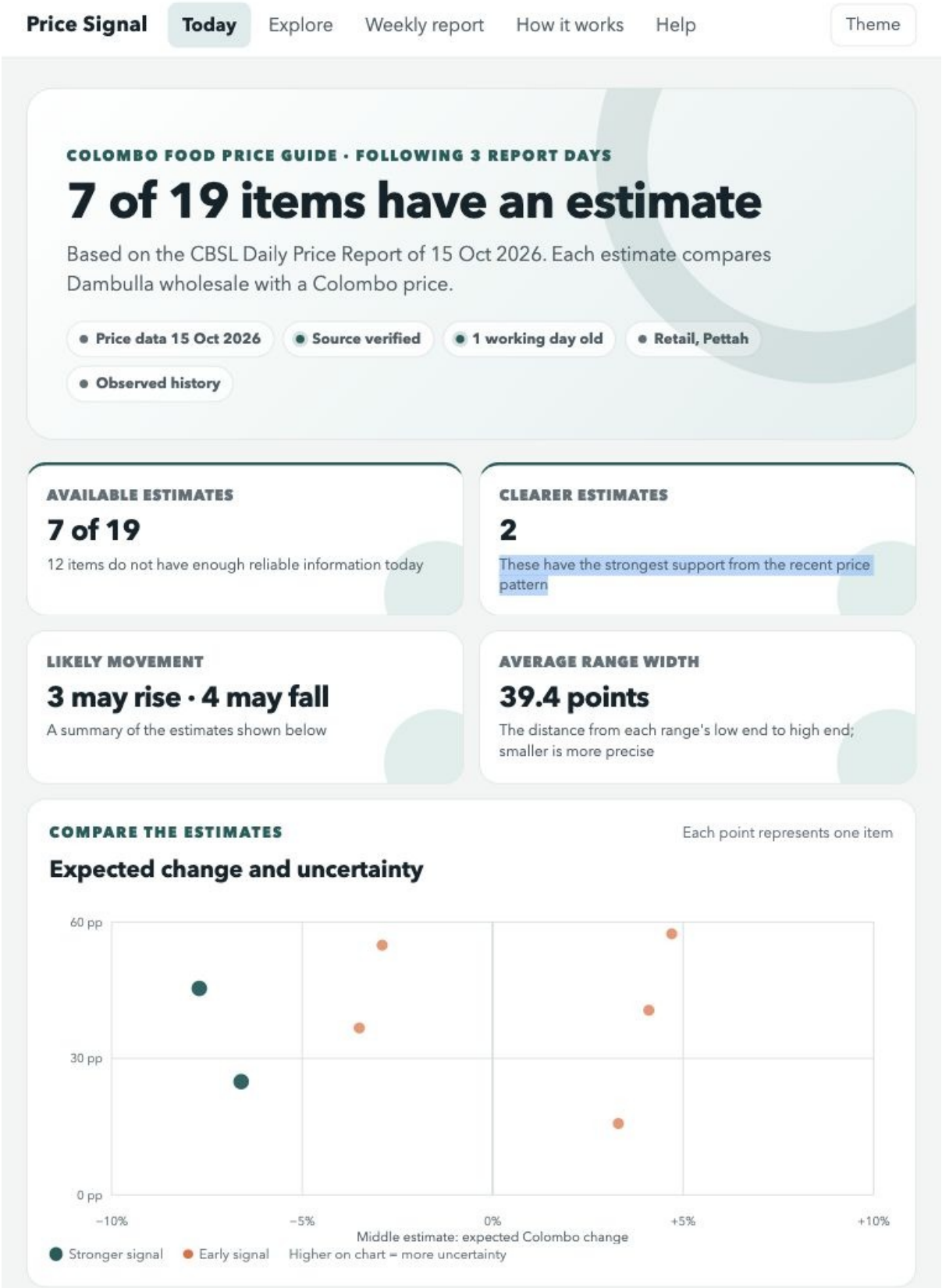


Figure 2. Current Today command centre using the 15 October 2026 report. It exposes availability, expected movement, uncertainty and source-health checks before detailed cards.

10.2 Explore

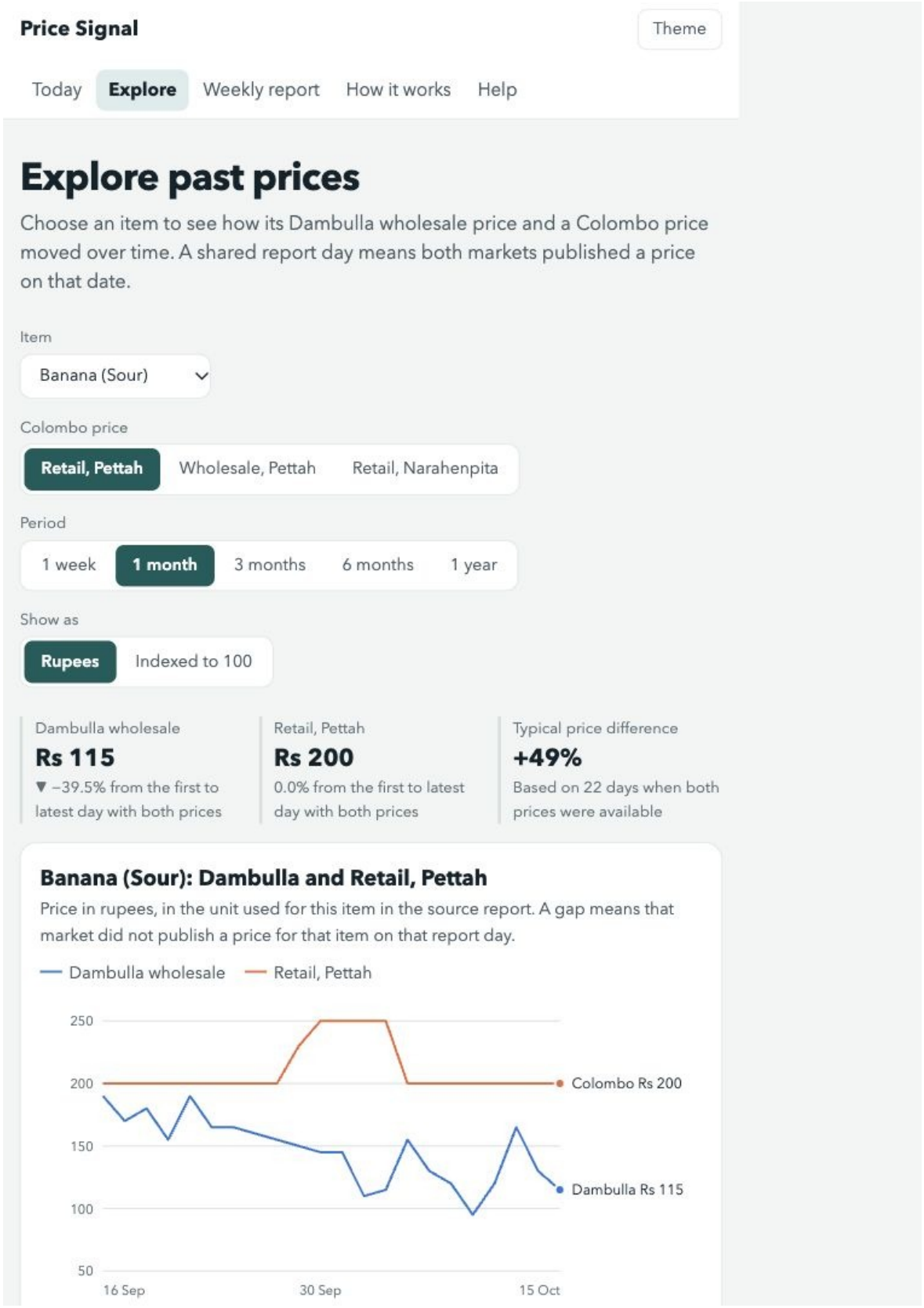


Figure 3. Current Explore page. Dambulla and Colombo series share dates; the price-difference section shows latest, typical and relative gaps in public-friendly language.

10.3 Weekly report

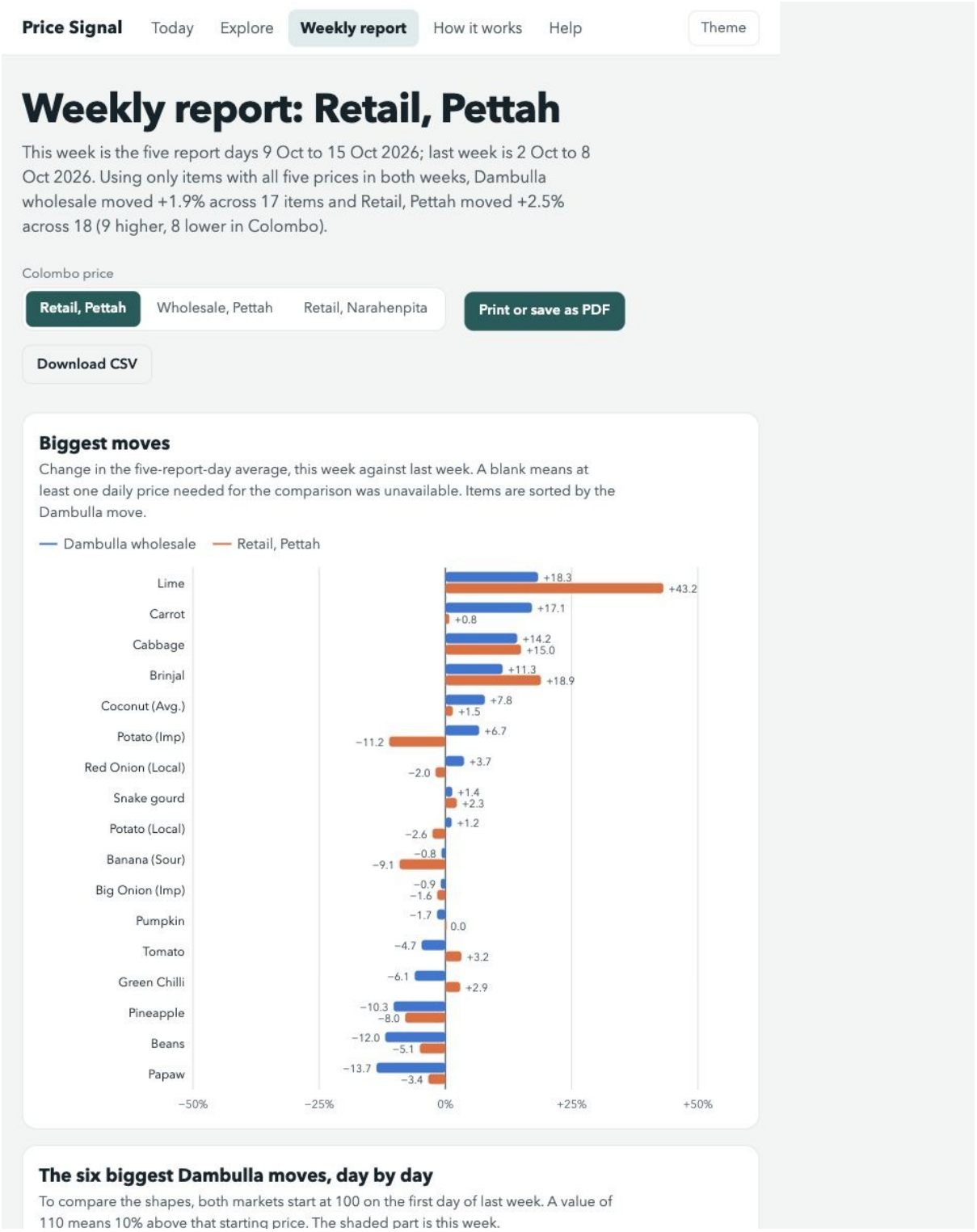


Figure 4. Current Weekly report. Only complete five-report-day windows are compared, with explanations for unavailable values and guarded export.

10.4 How it works

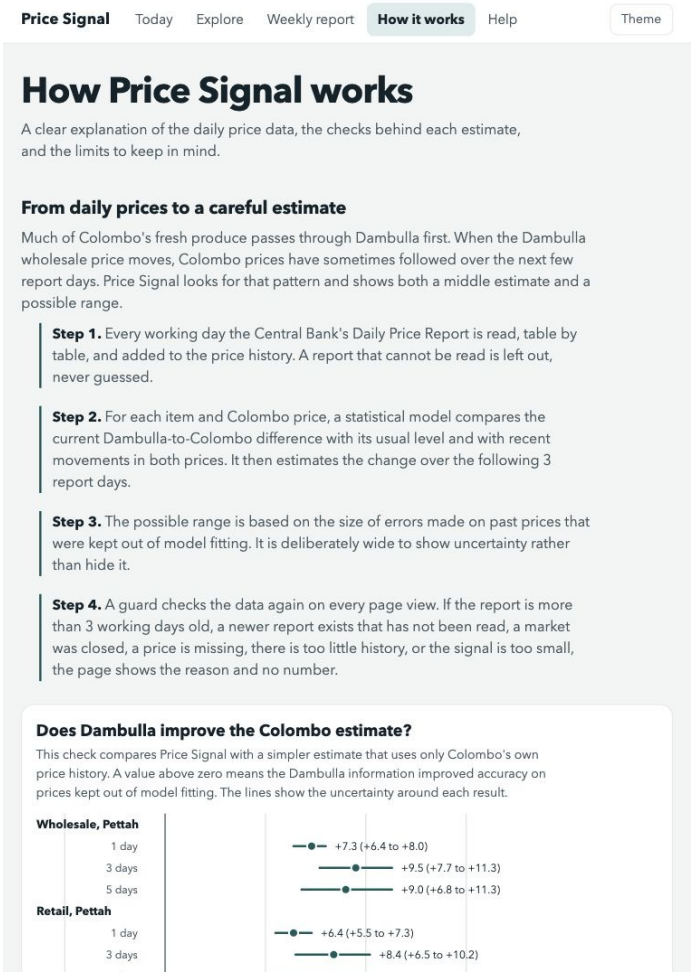


Figure 5. Current public evidence page, renamed from Method to How it works and revised for non-technical readers.

11. Reproducing the work

```
python3 -m pip install -r requirements.txt
python3 download_cbsl_reports.py
python3 extract_prices.py data/raw prices_all.csv
python3 robust_check.py prices_all.csv
python3 leadlag_check.py prices_all.csv
python3 decision_backtest.py prices_all.csv
python3 model.py prices_all.csv --evaluate
python3 challenger_check.py prices_all.csv
python3 model.py prices_all.csv
python3 export_recent.py prices_all.csv
python3 web/build_db.py
python3 web/server.py
python3 -m pytest -q
python3 refusal_report.py
python3 rolling_stress_check.py data/recent_prices.csv
```

12. AI assistance

Claude (Anthropic) and OpenAI Codex assisted with research, first drafts of code and documents, testing, review, safety hardening, the redacted permission evidence, interface work and report production. AI_USE.md records the file-level account. The author ran and reviewed the work and remains responsible for the system, evidence and submission.

13. Buyer evaluation

A small restaurant buyer participated on 16 October 2026; a vegetable-shop buyer and a hotel/catering buyer participated on 17 October. All permitted anonymous use of their responses.

Measure	Individual scores	Median
Main-card understanding	4, 3, 4	4/5
Trust as one buying input	4, 2, 4	4/5
Usefulness for work	5, 3, 4	4/5

All three completed Today, although one needed help. Two needed help or could not confidently explain the possible range. All three needed help or did not complete the historical comparison, and two questioned blank weekly values. The site now gives a plain-language range explanation, clearer shared-date language, the source-data date and explicit reasons for blank weekly values. This is formative evidence from three people, not proof of adoption or economic impact; the revised wording has not yet been retested.

14. Remaining submission actions

1. Complete the owner-authorized public deployment and verify /healthz, /api/status and all six public pages.
2. Add the exact public URL to the README, proposal, technical report and two-page summary, then regenerate the affected PDFs.
3. Record the three-minute demonstration from the same final commit and data date.
4. Complete the signed-out/private-browser rehearsal and preserve the checklist result.
5. Optionally retest the revised wording with one or two buyers; do not delay the submission if deployment and video deadlines are close.

Final status: the local implementation and evidence package are complete and verified. Public deployment, final URL insertion and the video remain the required completion gate.